What it reads before it answers, and what it builds

# Working with an AI agent in research, part 2

Johannes Wildfeuer · DTU Construct · DTU seminar on AI in research · 3 November 2026

# In twenty minutes




## What it reads

before you type a word

## The checks research needs

against the paper, before it leaves

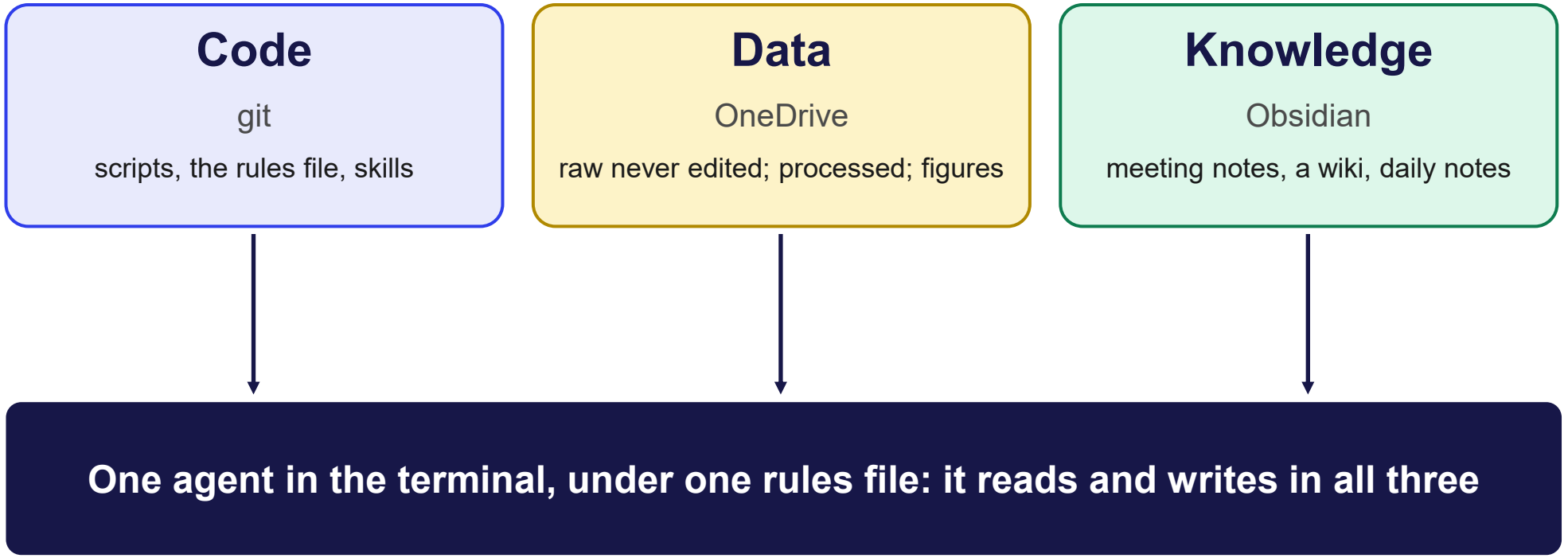
## What it builds

a tool from the knowledge base

**The agent is only as good as what it reads before it answers.**

**In a terminal, that is yours to choose, to check and to keep.**

# Where I start from




30 buildings · 5 data vendors · 3 people on one base

# 1 What the agent reads

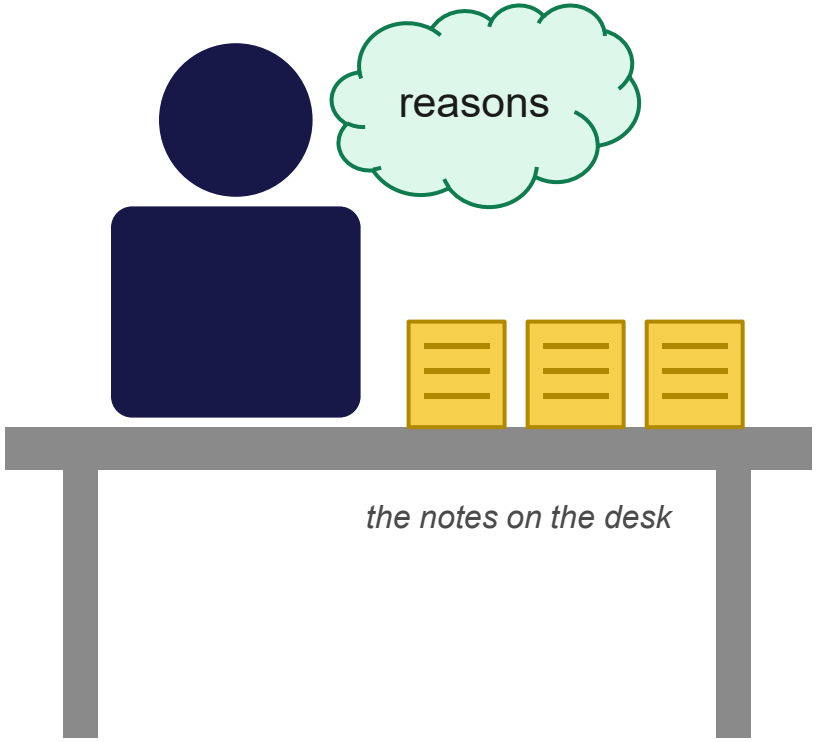
The colleague who forgets, the desk, tokens, and what it costs.

**What the agent reads**

The checks research needs

What it builds

# A brilliant colleague who forgets




**Reasons as well as yesterday**

**Forgets everything when the session ends**

**Knows only what is on the desk**

**The folder is the memory. The conversation is not.**

# The desk is laid before you type a word

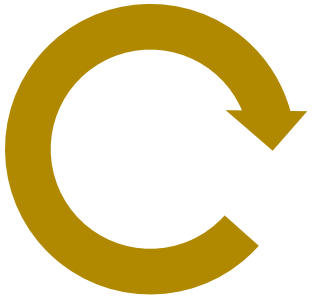
## Laid by rules

- rules file
- skills
- tools
- memory index

## Piles up

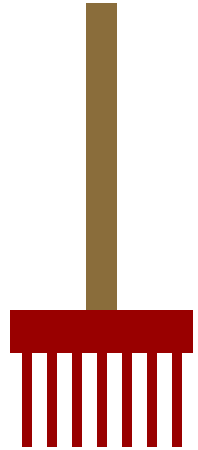
- what you say
- files it opens
- printouts
- its answers

## Read whole



reads all of it again, after every prompt

## Full: swept



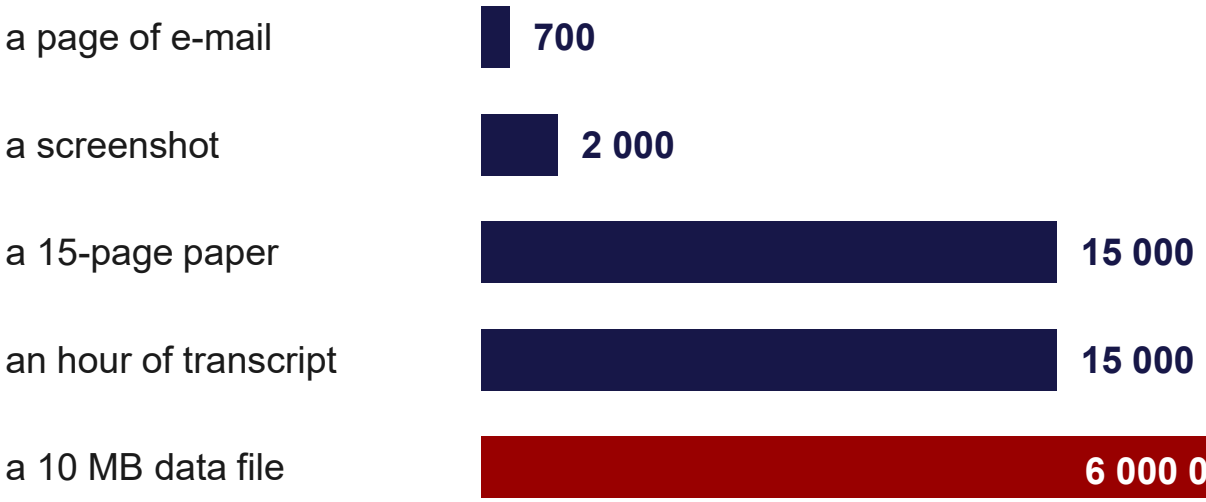
a summary kept, the rest forgotten

**No search in front of the model: what reaches it is what was laid out, all of it.**

# Tokens: the pieces everything is counted in

| Un | der | every | city | , | heat | is | travel | ling |
|---|---|---|---|---|---|---|---|---|

*about three letters each*

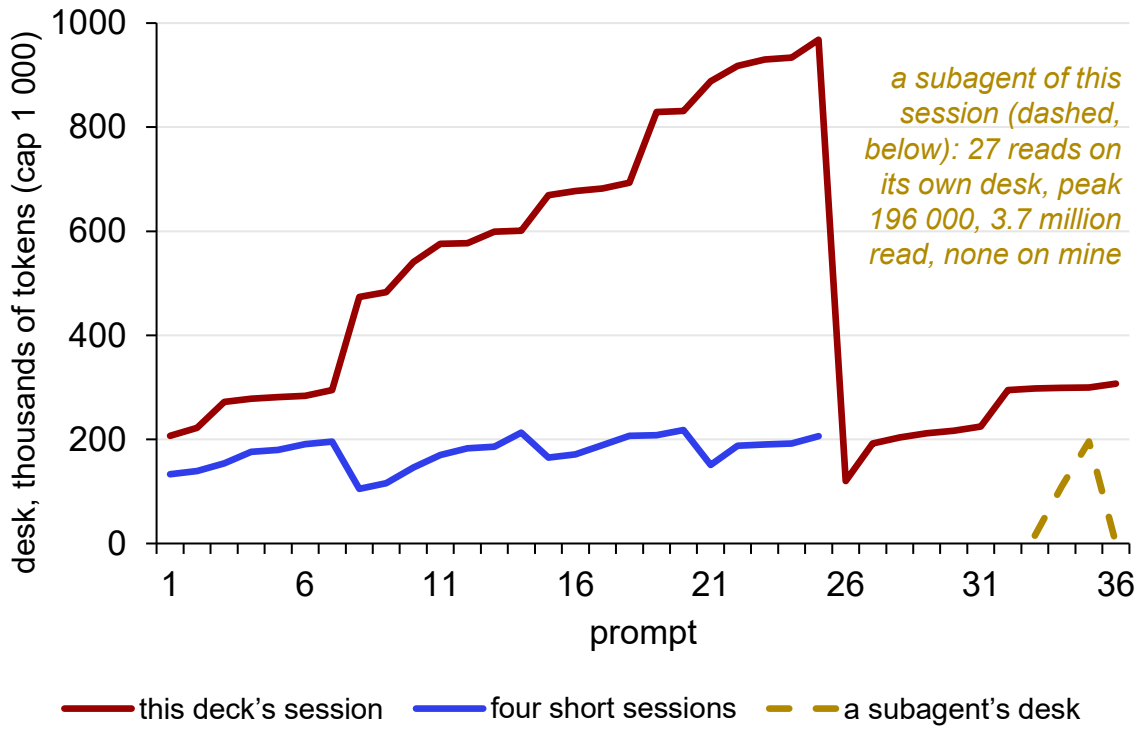



**500**

the same 10 MB file read by a script that prints twenty lines

**a “please”: 1 token**

# One long session, or several short ones?




*Measured from my session records, 96 sessions since September; across them a prompt causes about 4.5 reads of the desk, one per tool call. Red: the session that built this deck, 36 prompts that caused a read, 176 reads, 81 million tokens read, swept once at the cap. Blue: four short sessions, 25 prompts, 176 reads, 27 million.*

### Follow-ups: keep going

the pile is cached; a re-read is billed at about a tenth

### A new task: a new session

nothing in the old pile helps it; clear between tasks

### A long read: a subagent

its own desk, the dashed line: 3.7 million read there, nothing of it on mine

### Nothing is skipped

the desk is read at every tool call, 4.5 times per prompt: 81 million in one session

# My desk, measured

**60 000**

tokens before the first word

median of 73 sessions

about four papers' worth

**one week of use on the Pro subscription**



my session starts: about 3 %

with re-reads: about 15 %

*estimate at list prices, half the start from cache; sources in the notes*

## Small on the bill, large on the desk

the more on it, the less carefully each sheet is read; a long session carries the whole pile on every turn

## What cut it

skills typed, not offered

unused connected tools off

a rules file that routes to files read on a trigger

the agent's own notes: one screen per project, not a diary

# The rules file, and what does not belong in it

| Where | What goes in | When it is read |
|---|---|---|
| **The rules file** | who you are, where things live, the always-do rules, the command that checks the work; under 200 lines | *every session* |
| **Topic rules, context files** | procedures that matter for one part of the work | *on a trigger, or when matching files are opened* |
| **Skills** | a named procedure: how a release, a meeting note or a vault sweep is done | *its description every session; the text only when used* |
| **The agent's own memory** | what it learned from your corrections, one screen per project | *every session, the first 200 lines* |
| **Hooks and deny rules** | what must never or always happen: a folder it may not touch, a check before every commit | *enforced by the harness, not read by the model* |

**If you keep adding emphasis to a line in the rules file, it belongs in a hook.**

# Three of my skills, and a warning

## repo-sync

git housekeeping across every repository of the workspace: commit, push, reap merged branches and stale working copies

## vault-upkeep

the recurring sweep of the knowledge base: my comments on the task board, garbled names, the inbox

## restructure-context

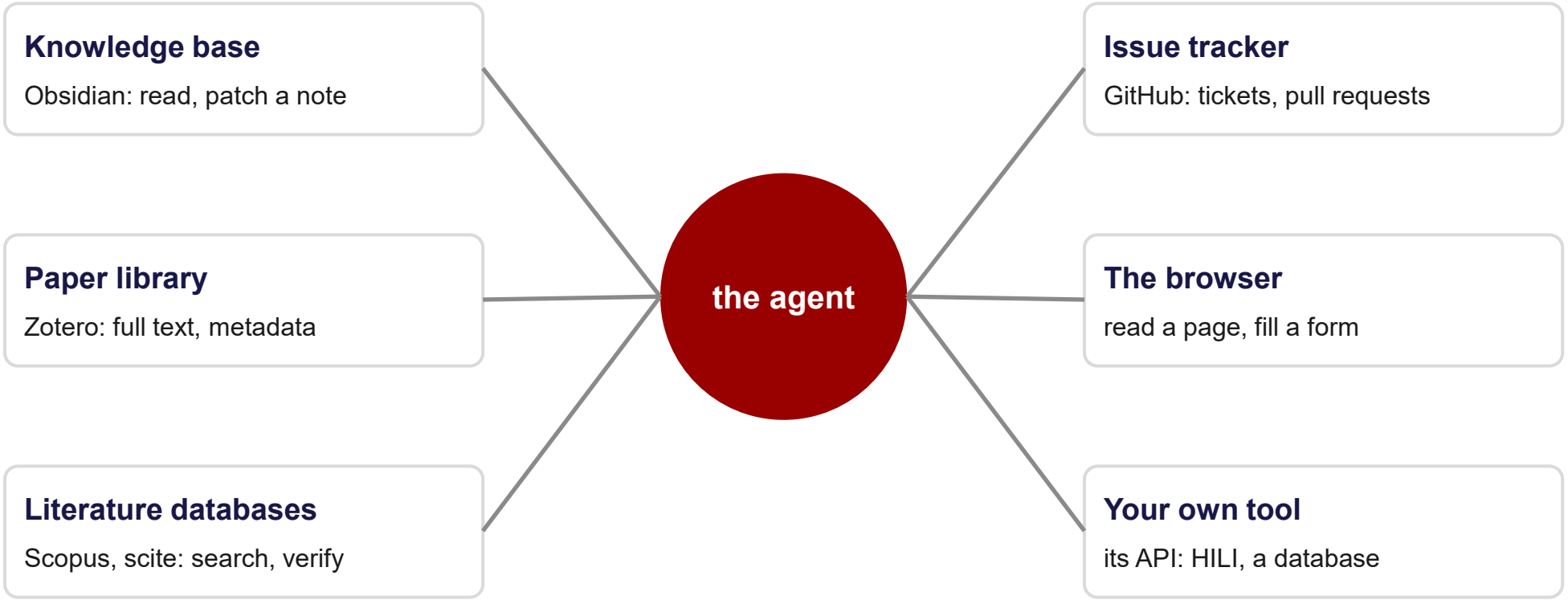
the audit of the rules files, context files and memory: duplicates, contradictions, stale lines, the size of the desk

## These files grow and go stale without anyone noticing

my rules file was 22 KB and the agent's memory 190 KB before the audit of October. Review them; the agent can run the review itself and hand you a decision table to mark

**A procedure typed once; an audit run by the agent; a table marked by you.**

# MCP: the agent's hands




**Model Context Protocol: one plug standard; any tool can offer the agent named actions.**

# 2

# The checks research needs

The paper itself, a referee before anything leaves, and three stores for three jobs.

# Open the paper itself before a claim carries weight

### A note is a summary

written for an earlier question; what it omits, it says nothing about

### Open the full text

through the library; find the page and the equation

### Record the check

the note, or the paper, and where

### Models mostly get the reference right

what the paper says is where they go wrong: with tools, one attributed claim in four is not on the cited page

### 18 October 2026

a method in the paper, absent from its note; two claims built on abstracts fell when the full texts were read

**A summary's silence is not evidence. Only the paper is.**

# The agent as referee, not only as writer

1

## Devil's advocate

a document before it leaves: every claim against its source

2

## Council

a contested decision: independent answers, blind review, one synthesis

3

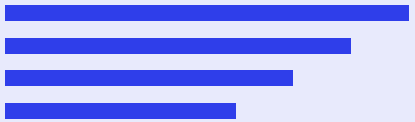
## Citation audit

each reference found in the library or struck

**A separate agent on the stronger model, reading the source, not the summary.**

**It never edits on its own; I decide every fix.**

# Three stores, three jobs



## The board

**a task is a line**

one line, one place; a date pulls it into the day



## The day

**a day is a log**

capture, and what happened; never moved



## The ticket

**a decision is a thread**

a question too big for one session; I answer, it closes

**The agent refines, rolls, charts and recommends.**
**It never puts a task on my day, and never decides a ticket.**

*Next rule: a task rolled five times goes to the decision list, not to tomorrow.*

# 3 What it builds

A tool from the base, and what to do when it goes wrong.

What the agent reads
The checks research needs
**What it builds**

# HILI: from the base and a sketch to a tool




*the trailer, two minutes · click the picture to play*

**7 weeks**

from the first prototype to version 1.27 (part-time)

**33 versions**

tagged since 19 September

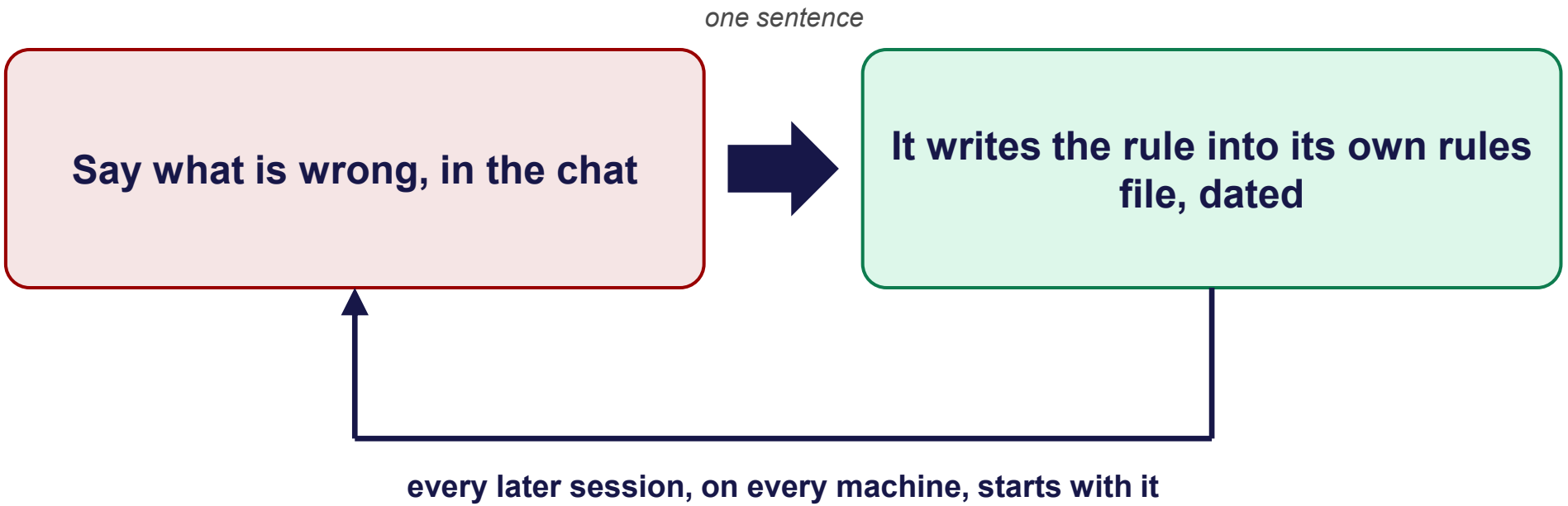
**0 lines**

of its code typed by me

**What I put in**

the data model, two sketches, verdict tables, one-line tickets

# When it goes wrong, say so




**A correction told once becomes a rule for every session. The setup improves through you.**

# What this half added

1. The desk: what the agent reads before the first word is laid by you, in files
2. Its cost, measured: small on the bill, large on the desk
3. The paper itself before a claim carries weight; the agent as referee, never the last word
4. Three stores for three jobs: a line, a log, a thread
5. A tool built from the base and two sketches, and a film about it in a night

Johannes Wildfeuer · jniwi@dtu.dk · these slides were made with an agent from my knowledge base


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